



Survey on Machine Learning and Deep Learning Applications in Breast Cancer Diagnosis

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Abstract

Cancer is a fatal disease caused due to the undesirable spread of cells. Breast carcinoma is the most invasive tumors and is the main reason for cancer deaths in females. Therefore, early diagnosis and prognosis have become necessary to increase survivability and reduce death rates in the long run. New artificial intelligence technologies are assisting radiologists in medical image scrutiny, thereby improving cancer patients' status. This survey enrolls peer-reviewed, newly developed computer-aided diagnosis (CAD) systems implementing machine learning (ML) and deep learning (DL) techniques for diagnosing breast carcinoma, compares them with previously established methods, and provides technical details with the pros and cons for each model. We also discuss some open issues, research gaps, and future research directions for the advanced CAD models in medical image analysis. Over the past decade, machine learning and deep learning have emerged as a subfield of artificial intelligence (AI), whose healthcare industry applications have provided excellent results with reduced cost and improved efficiency. This survey analyzes different classifiers of machine learning and deep learning approaches for breast cancer diagnosis. Results from previous studies proved that deep learning outperforms conventional machine learning for diagnosing breast carcinoma when the dataset is broad. Research gaps from the recent studies depict that practical and scientific research is an urgent necessity for improving healthcare in the long run.

Keywords Breast cancer · Machine learning · Deep learning · Convolution neural network (CNN) · Computer-aided diagnosis (CAD)

Introduction

Our bodies have millions of cells. Cancer starts when cellular changes cause these cells to grow excessively, creating a lump called a primary tumor [1]. According to data reported in Cancer Statistics, 2018 [2], approximately 2.25 million people live with this disease. New registered cancer patients exceed 1,157,294, and nearly 784,821 cancer-related deaths are reported annually. The risk of dying from cancer is approximately 7.34% in males and 6.28% in females. In

males, 25% of cancer deaths were reported for oral cavity and lung cancer, whereas breast and oral cavity cancer accounts for 25% of cancers in females [2].

Figure 1 shows a detailed description of cancer statistics for the year 2018. As we can see, breast cancer is the most frequent carcinoma and constitutes 14% of all cancers in women. Globocan 2018 study [2] reported approximately 162,468 new breast cancer cases and 87,090 deaths.

Incidences of breast carcinoma are increasing at a startling rate in India, particularly in younger age groups—30s and 40s (Fig. 2). Women aged 20–59 years are most often affected by breast cancer [4]. In India, detection occurs very late, which is the main reason for low survival rates for breast carcinoma [5]. Over the last few decades, cancer-related research is being performed. Scientists have applied different methods to predict disease before they cause symptoms. Therefore, accurate diagnosis and prognosis of breast cancer are regarded as exciting and challenging tasks for physicians in the medical and healthcare communities [6].

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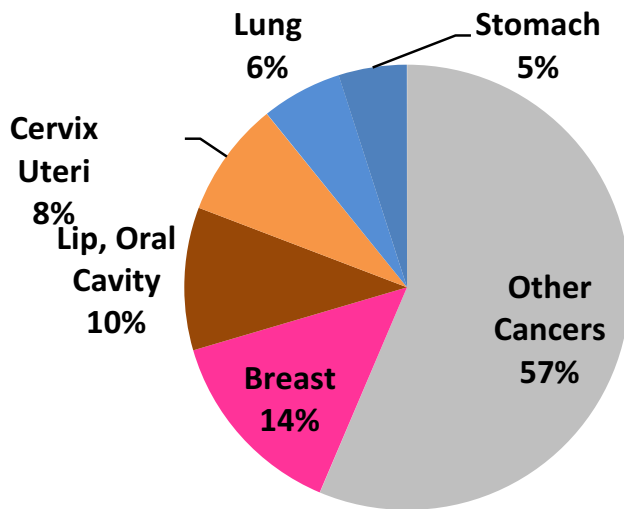


Fig. 1 Cancer Statistics 2018[3]

Manual diagnosis of breast cancer is tedious and time-consuming; thus, there arises a need for automatic diagnosis. The current healthcare systems have proven to be helpful, but they are prone to errors. Therefore, medical image classification using CAD has emerged as an efficient tool that can help doctors classify medical images in different categories, leading to early diagnosis and treatment [8–10]. In this context, ML and DL have come up with algorithms capable of more accurately diagnosing the disease at an earlier stage and thus diminishing the number of readmissions in hospitals and clinics. AI techniques can thus boost the procurement of new fidelity protocols in medicine and reduce healthcare costs due to misdiagnosis [11].

This survey focuses on recent research developments for early diagnosis of breast carcinoma using ML & DL techniques. It highlights the strengths and weaknesses of various research methods distinguishing this survey paper

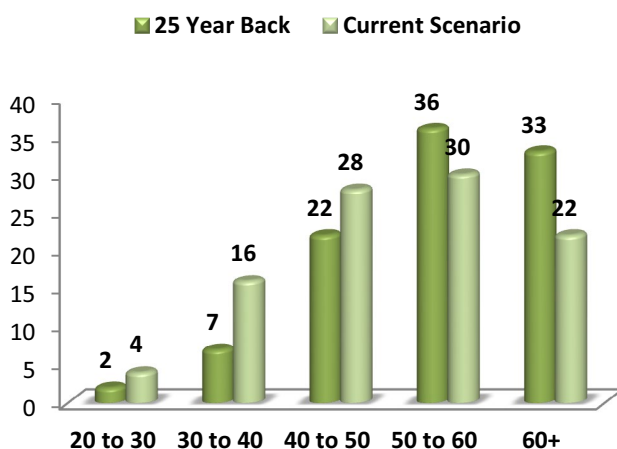


Fig. 2 Incidences of breast cancer in India [7]

from other existing surveys. Challenges and future directions are also discussed that would guide a new researcher in this field.

The paper is structured in the following way: “Breast Cancer” introduces breast cancer with its categorization, evolution stages, imaging modalities, and performance measures. “Computer-Aided Diagnosis [18]” focuses on medical image analysis using CAD and outlines machine learning and deep learning-based CAD models. Literature survey for diagnosing breast carcinoma is summarized in “Literature Survey”. “Research Gaps and Future Directions” discusses research gaps and future research recommendations for the advanced CAD system in the field. Finally, the conclusion is outlined in “Conclusion”.

Breast Cancer

Lobules (i.e., milk-producing glands) or the ducts (i.e., canals for carrying the milk from the lobules to the mamilla) of the breast are the primary regions where breast cancer initiates. The unbounded evolution of cancer cells can spread to other robust breast tissue and transit to the arms’ lymph glands. The lymphatic nodes are responsible for moving the tumor cells to further body parts.

Physical signs of breast carcinoma include breast tenderness, irritation, nipple discharge besides breast milk, inverted nipple, change in breast size, soreness, or blister under your arm. Risk factors associated with breast cancer may include age, overweight, low physical activity, excessive alcohol consumption, dense breast, genes, early menstruation, and advanced maternal age [12]. S. Simsek et al. [13] evaluated the impact of several factors that contribute to breast cancer at various time intervals (1 year, 5 years, and 10 years). The authors used two algorithms, artificial neural network (ANN) and logistic regression (LR), and also show how the significance of such parameters changes over time. Researchers concluded that the factors that are considered relevant for cancer at an initial stage might not contribute equally to predicting patients’ survivability over the long term. For example, age is regarded as an essential factor for forecasting survivability, but its potentiality drops with time, maybe due to the empowerment of other factors such as grade, lymph nodes positive, etc. Therefore, the importance of all such elements must be considered when developing a model for predicting breast cancer survivability in the future.

Categorization

Breast carcinoma is investigated by distinguishing tumors as malignant and benign. The medical practitioner and pathologists need a reliable diagnostic procedure to differentiate these two classes of tumors [14].

It can be categorized into two classes: invasive and non-invasive/in situ. Invasive cancer starts in the milk ducts (canals) and can spread to the surrounding breast tissue and other breast parts, whereas non-invasive cancer cannot escalate within or to other body parts.

Invasive and non-invasive breast carcinomas are further categorized as follows:

1. **Ductal carcinoma in situ:** It is non-invasive carcinoma where cancer cells have not transmitted in the neighboring breast tissue and are limited to the breast ducts.
2. **Lobular carcinoma in situ:** In this carcinoma also, cancer cells have not infected nearby tissues, but cells spread in the milk-producing glands of the breast.
3. **Invasive ductal carcinoma:** This is the most prevalent category of cancer. It initiates in milk ducts of the breast and can then spread to neighboring tissues of the breast. This might extend to adjacent tissues in the body once it starts spreading beyond the breast ducts.
4. **Invasive lobular carcinoma:** This cancer starts in lobules and then infects adjacent tissues.

Some less frequent categories of breast cancer include:

1. **Phyllodes tumor:** This category of breast cancer is infrequent and mostly benign but can be malignant too. It can grow in the connected tissues of the breast.
2. **Angiosarcoma:** Blood vessels or lymph vessels are the primary sources where this carcinoma grows in the breast.
3. **Inflammatory breast carcinoma:** It is the rarest of a rare case and accounts for 1 to 5% of all cases. Having this carcinoma causes the breast to swell, felt very hot, and becomes red.
4. **Triple-negative breast carcinoma:** It accounts for 10 to 20% of all cases. This type of carcinoma occurs when a tumor does not have extra human epidermal growth factor receptor (HER2) proteins on its surface and lacks estrogen and progesterone receptors. Hormonal therapy for breast cancer is not sufficient, so this type of carcinoma is difficult to treat.

Evolution Stages

Stages of breast carcinoma can be divided based on the tumor size and how much it has grown in the nearby tissues. Table 1 shows the categorization of breast cancer in four main stages [15].

Imaging Modalities

Different modalities can be used to diagnose breast carcinoma. The following are some of the commonly used modalities used by the researchers in recent years:

1. Screen Film Mammography (SFM)[16, 17]

SFM shows an X-ray picture of the breast.

Pros:

- Detects cancer at an early stage
- Standard modality
- High sensitivity for breast with fatty tissue
- Cost-effective

Cons:

- Low sensitivity with dense breast
- Not digital

2. Full field digital mammography (FFDM)/digital mammography (DM) [16, 17]

Here, X-ray films are converted into mammographic photographs of the breast (digital images) by digital detectors.

Pros:

- Detects cancer at an early stage
- Effective and standard modality
- Reduce cancer rate by 15%

Cons:

Table 1 Breast carcinoma evolution stages

Stages	Description
Stage 0	Abnormal cells are present but not grown in neighboring tissues
Stage 1: In situ	Cancer cells are confined only in the lobules and ducts of the breast
Stage 2: Localized	Some lymph nodes are involved, and the size of the tumor is confined between 20 and 50 mm
Stage 3: Regional	Tumor size is greater than 50 mm, and cancer grows in skin or chest wall with more involvement of lymph nodes
Stage 4: Distant	The tumor can spread further to any part of the body and can be of any size

- High false-positive and false-negative findings
- High radiation exposure
- High cost compared with SFM
- Low specificity thus leads to unnecessary biopsies

3. Digital breast tomosynthesis (DBT)/3D mammography [16, 17]

Similar to mammogram, here, breast images are acquired at multiple angles and thus create 2D and 3D-like pictures of the breasts.

Pros:

- Detects cancer in women with dense breast
- Reduce no. of false positive and false negative
- Multiple 3D images with a single screening

Cons:

- Expensive
- Average radiation dose is 1 to 2 times higher than DM

4. Ultrasound (US) [16, 17]

US images are also called sonograms. It uses high-frequency sound waves.

Pros:

- No radiation involved
- It can be used for pregnant women and women with dense breast
- It is the best way to figure out whether the anomaly is solid or fluid-filled.
- Safe, accurate, and low-cost

Cons:

- Operator dependent
- Many false-positive and false-negative results

5. Magnetic resource imaging (MRI) [16, 17]

Another modality to diagnose breast carcinoma is MRI. Here, powerful magnetization and radio signals generate high-quality pictures of the breast.

Pros:

- Used for high-risk patients
- High sensitivity as compared with the US

- Identifies incredulous areas that can further be used for biopsy

Cons:

- Expensive compared with US and DM
- Low specificity
- Risk of a false-positive result
- Body temperature may increase during long MRI

6. Histopathology (HP) images [16, 17]

These are Hematoxylin & Eosin-stained images. In this, samples are taken from abnormal breast regions (called a biopsy) and are observed under the microscope.

Pros:

- Diagnose different breeds of cancer instead of detecting only malignancy because of multi-color images
- A comprehensive study of tissues is possible
- Fewer chances of missing cancer in the initial stages because multiple region-of-interest (ROI) images are created from whole slide images

Cons:

- High proficiency is required as the manual analysis is tedious and time-consuming
- Color variation and different staining procedures can cause misdiagnosis

Performance Measures

The confusion metric is one of the estimates used for evaluating the performance of a model. It indicates the relationship between predicted and actual values (Table 2).

True positive (TP)—patients diagnosed positive and are genuinely positive.

True negative (TN)—patients diagnosed negative and are actually negative.

False positive (FP)—patients diagnosed positive but are truly negative.

False negative (FN)—patients diagnosed negative and are genuinely positive.

Other metrics for evaluating the effectiveness of a model are discussed below:

Table 2 Confusion matrix

	Predicted positive	Predicted negative
Actual positive	True positive	False negative
Actual negative	False positive	True negative

1. Accuracy: It is the ratio of correctly categorized samples to the total number of samples and is interpreted as:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

2. Sensitivity/recall/true positive rate (TPR): It evaluates what percentage of patients are correctly identified to have a particular disease (+ve instances) and illustrated as:

$$\text{Sensitivity (recall)} = \frac{TP}{TP+FN}$$

3. Specificity/true negative rate (TNR): Determines what percentages of patients are correctly identified not to have a particular disease (–ve instances) and elucidated as:

$$\text{Specificity} = \frac{TN}{TN+FP}$$

4. Precision/positive predicted value (PPV): Specifies what percentage of patients are actually relevant (+ve) and described as:

$$\text{Precision} = \frac{TP}{TP+FP}$$

5. F-measure: Interpreted as the harmonic mean of precision and recall and is defined as:

$$\text{F-measure} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

6. False-positive rate (FPR): It predicts a positive value when the result is negative. Thus, FPR is 1-specificity and is given as:

$$\text{FPR} = \frac{FP}{TN+FP}$$

7. Area under the receiver operating characteristic curve (AUC-ROC): It defines a model's capability to distinguish between different classes. It is drawn by plotting sensitivity on the y-axis (true positive rate) and 1-specificity on the x-axis (false-positive rate). Thus, model performance is evaluated by calculating the area under this curve, whose value lies from 0 to 1. The higher the value, the more powerful and reliable the model is at distinguishing positive patients as positive and negative patients as negative.

Computer-Aided Diagnosis [18]

Earlier, medical image processing was done manually. Still, it faces three main issues—firstly, the unavailability of more than one pathologist in one place. Secondly, manual image analysis is a very laborious and stressful operation,

and thirdly, the professional experience and domain knowledge of pathologists are very crucial for breast carcinoma diagnosis. Thus, to overcome misdiagnosis in the early stages, CAD has emerged, which acts as a “second notion” for solving breast cancer multi-classification problems [16].

CAD systems require processing on medical images; thus, extensive computational algorithms must be developed to process those images. CAD is reasonably good at detecting invasive breast cancers in their early stages but increases the peril of false positive results. Thus, accuracy and workflow efficiency must be considered before implementing CAD in clinical practice, and secondly, to avoid improper use, user training is equally important to understand the limitations and capabilities of CAD systems (Fig. 3) [19].

CAD systems are categorized into two categories [21]:

Computer-aided detection (CADE) systems: Provides computerized detection and localization of anomalies in medical images

Computer-aided diagnosis (CADx) systems: Serve to determine the class of anomalies to classify them as benign or malignant. It can be used as a step further to the detection model, to provide additional information and further help in decision making.

CAD Using Machine Learning

CAD system implementing machine learning comprises of the following stages:

- (1) Image preprocessing, i.e., removing the noise and thus cleaning the image and preparing it for the subsequent stage [22]

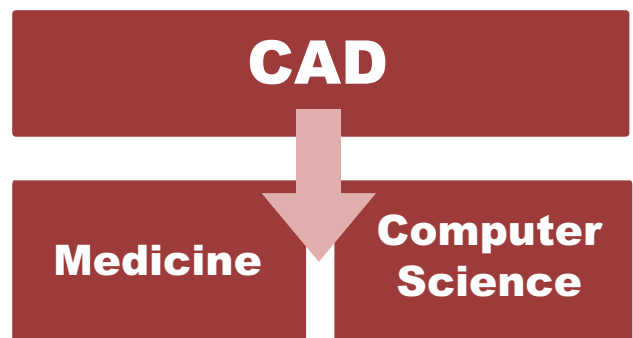


Fig. 3 CAD acts as a mediator connecting medicine with computer science [20]

- (2) Image segmentation, i.e., partitioning the image into several parts, to work only on the segmented regions called regions of interest (ROI) [22]
- (3) Feature extraction and selection, i.e., extracting relevant features from the pre-processed images. Feature selection provides many advantages that include cost reduction, lesser learning time, and increased accuracy [23]
- (4) Classification, i.e., assigning labels or classes to different groups using several classifiers such as support vector machine (SVM), K-nearest neighbor (KNN), Naïve Bayes (NB), decision tree (DT), random forest (RF), logistic regression (LR), etc.

CAD Using Deep Learning

Before classifying the entire image, neural networks are competent in learning transitional, hierarchical, and more intellectual characterization of the data [24], and thus, deep learning is categorized as a representation learning technique. Deep learning learns characteristics directly from the data using several non-linear computation layers [9]. Contemporary research communicated that “convolutional

neural network (CNN)” attains remarkable performance in cancer detection and diagnosis [25]. Layers adjoining the input layer learn low-level features and are more generic. In contrast, layers adjoining the output layer learn distinctive features of the input image and, thus, are more specific [26]. Figure 4 shows the ML- and DL-based CAD systems. CAD system based on CNN implements the following layers:

- (1) Convolutional layers for performing convolution operation that automatically extract features from the image
- (2) Pooling layer for reducing the dimensionality of features
- (3) Fully connected and Softmax layer for performing classification.

When inadequate data is available to train a generalizable model, data augmentation is executed for enlarging statistics of data points. Operations performed for augmentation involve rotation, noise inclusion, image cropping, and other geometrical activities [27].

Two ways to implement CNN are

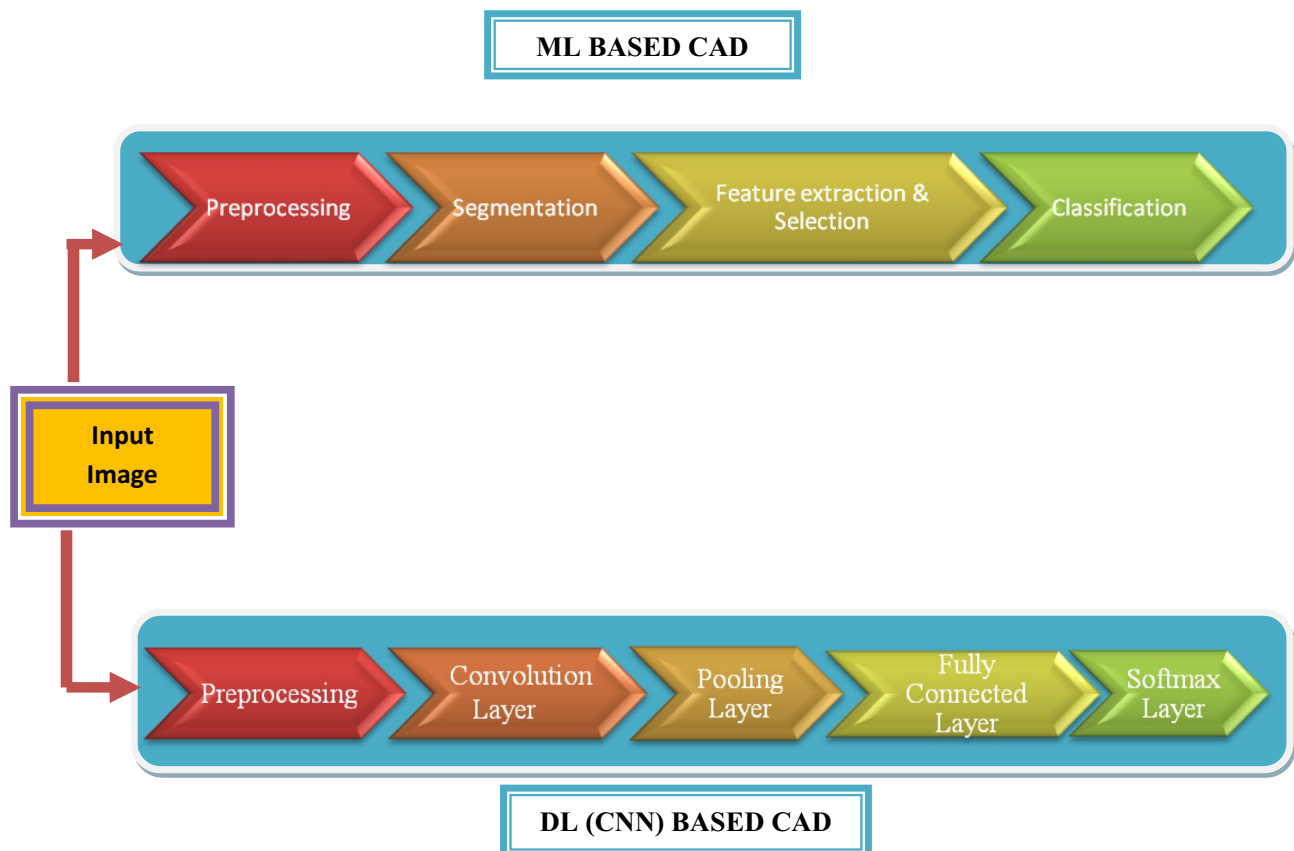


Fig. 4 ML-based and DL (CNN)-based CAD systems

- (1) Training from scratch: CNN can be trained from scratch by providing enough training examples as input. When creating CNN from scratch, considerable skill and experience are required. Selecting suitable hyperparameters such as the learning rate, the size of convolutional filters, the number of layers, etc., is challenging. Thus, the entire process is time-consuming and is usually implemented on powerful GPUs with high processing power.[28].
- (2) Transfer learning: Training a CNN on a small database affects the model's generalizability and usually results in overfitting. Transfer learning involves transferring the knowledge from the source domain, which has extensive samples of data, to the target domain, where data is comparably significantly less for training a deep CNN. Pretrained models, i.e., models trained on non-medical images, which usually have large datasets, can prepare the models for medical images with smaller datasets. Pretrained networks can be used in three approaches, which are discussed below:
 - As the baseline model: This approach exploits only the architecture of the pre-trained network. Thus, the complete model is built from scratch by random initialization of weights. [29].
 - As feature extractor: Here, convolutional base is frozen, and the pre-trained weights remain unaltered, i.e., the output from the convolutional base is fed directly to the classifier [29]. The pre-trained model is used to excerpt the essential features from the input dataset.
 - Fine tuning: Through this method, weights are passed into the intended network from a pre-trained network.

This can be implemented in two ways: fine-tuning the system—layer-wise—and secondly, performing partial training of the network [29].

Literature Survey

Papers Reviewed

This survey explores 81 peer-reviewed papers, of which 23 articles diagnose breast carcinoma using classification algorithms of machine learning while the rest implements deep learning techniques. Figures 5 and 6 show the statistical distribution of ML and DL articles and imaging modalities used in the survey.

Databanks

The list of publicly available datasets used by the researchers in recent years is given in Table 3

Electronic Repositories

This survey paper analyzes several studies on machine learning and deep learning classifiers to diagnose and classify breast carcinoma. Searching criteria for inclusion and exclusion of the peer-reviewed journal paper are based on the following keywords: breast cancer, computer-aided diagnosis, machine learning, deep learning, classification, detection, computer-assisted diagnosis, and computer-aided detection. The researchers' electronic databases used in the survey include Elsevier, Springer, IEEE, Wiley, Oxford, Plos One, Hindwai, etc.

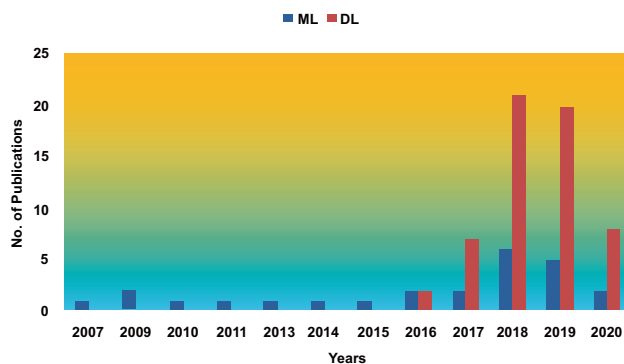


Fig. 5 A statistical distribution of ML and DL papers analyzed in the survey

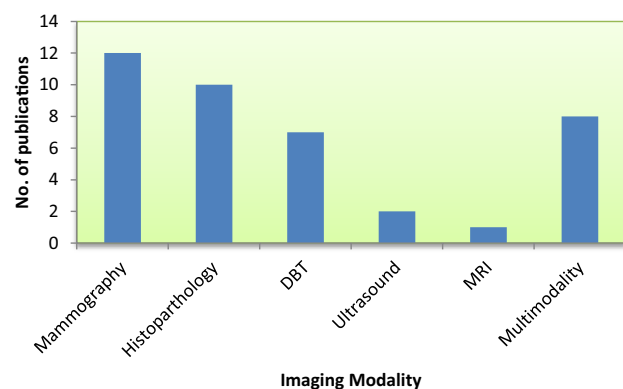


Fig. 6 A statistical distribution of imaging modalities used in the survey

Table 3 Databanks

Digital repository	Dataset name	Dataset description
UCIMLR/University of California Irvine Machine Learning Repository [30]	WBCD/Wisconsin Breast Cancer Dataset	WBCD is a commonly used dataset among researchers working on machine learning research. It contains 569 instances with 32 attributes
BCDR/Breast Cancer Digital Repository[31]	(a) BCDR-FM/ Film Based Digital Repository (b) BCDR-DM/ Full Field Digital Mammography based Digital Repository	It comprises mammography and ultrasound images of 1734 patients(1010 BCDR-FM and 724 BCDR-DM) along with their clinical history
MIAS /Mammography Image Analysis Society[32]	MIAS/Mammography Image Analysis Society	It has 322 digital mammography breast images at 50-micron resolution
Mini-MIAS[33]	Mini-MIAS	Here, the resolution of mammographic images (MIAS) has been reduced to 200 microns, and the image size is 1024×1024 pixels
The University of South Florida [34]	DDSM/Digital Database for Screening Mammography	It comprises 2620 cases which are annotated in 3 categories—normal, benign, malignant
BreakHis [35]	BreakHis/Breast Cancer Histopathological Database	It consists of a total of 7909 microscopic images of breast tissue collected at different magnifying factors. 7909 cases are divided into 2480 benign samples and 5429 malignant samples
SEER[36]	SEER/ Surveillance Epidemiology and End Result	SEER dataset provides stage-wise survivability statistics of breast cancer data
InBreast [37]	InBreast	This database consists of 115 cases with 410 images
IRMA [38]	IRMA/Image Retrieval in Medical Applications	It consists of 10,509 reference breast images

Comparative Research Using Classification Algorithms of Machine Learning

This section presents a comparative study from various authors implementing diverse machine learning classifiers for diagnosing breast cancer.

Osareh and Shadgar [39] categorized benign and malignant tumors on fine needle aspirates of breast lumps (FNAB) and Gene Microarray datasets. The authors performed feature extraction using principal component analysis (PCA), feature ranking using signal to noise ratio (SNR), and feature selection using a wrapper-based approach, i.e., sequential forward selection (SFS). Finally, classification was done using three classifiers—SVM, KNN, and probabilistic neural network (PNN). The authors concluded that the best results could be obtained by appropriate selection of kernels and classifiers. SVM with the RBF kernel attains the accuracy of 98.80% on FNAB and 96.33% on the microarray dataset after applying SFS and PCA. Their work suggested that SVM models with radial basis function (RBF) kernel could be further explored as a diagnostic aid for breast carcinoma diagnosis.

Feature selection and reduction are very crucial parameters that affect classifier performance. The finest classifier will perform worse if features are not chosen wisely. Karabatak

and Ince [40] used association rules (AR) on WBCD and reduced feature dimensions from nine to four. The authors obtained 95.6% and 97.4% accuracy with 4-inputs and 8-inputs, respectively, by performing classification using neural network (NN). In contrast, the rough set (RS) reduction algorithm eliminates redundant attributes and identifies essential features relevant to the model [41]. Chen et al. [41] implemented the RS algorithm on WBCD along with the SVM classifier and obtained 99.41% accuracy for a subset with five critical features. K. Goyal et al. [1] implemented seven different algorithms—Bagging, Ada Boost, LR, SVM, RF, NB, and DT on WBCD. The authors compared classifiers' results by implementing various feature selection techniques such as PCA, subset evaluation using best first, and correlation attribute evaluation. Ada Boost and LR classifier attained significant performance using PCA with an accuracy of 97.92% for the reduced dataset.

Similarly, Shaikh et al. [42] implemented four classifiers, namely, NB, KNN, J48, and SVM, on WBCD and BCDR. 'WrapperSubsetEval,' the wrapper feature selection technique provided by the Weka tool, was applied to both datasets. Results show that the best results were given by SVM with an accuracy of 97.91% and 95.10% on WBCD and BCDR. Some of the well-known research on breast carcinoma diagnosis is summarized in Table 4.

Table 4 Summarization of distinguished work on breast cancer diagnosis using machine learning classifiers

Data set used	Method adopted	Accuracy achieved	Pros	Cons	Reference
UCI machine learning repository	Feature selection: <i>genetic algorithm (GA)</i> Classification: <i>SVM</i>	Small scale—GA + linear SVM <i>bagging</i> : 98% GA + RBF SVM <i>boosting</i> : 98.28% Large scale—RBF + SVM <i>boosting</i> : 99.52%	- SVM ensemble model performs better than single SVM classifier - Feature reduction using GA reduces computation time to a large extent in small datasets	SVM ensemble model based on boosting took a long time in construction	[12]
2301 patients (clinical data records)	Support vector regression ((SVR) with RBF, linear and polynomial kernel, decision tree regression (DTR) model, stochastic gradient descent model	Best accuracy—SVR-linear and DTR	- SVR linear model and DTR model best suited for survival time prediction - Tumor integrated clinical feature(TICF) performs better than the Nottingham prognostic index (NPI)	-	[14]
SEER	NB, LR, DT	Survivability prediction on all stages together can cause overestimation of performance	The stage-specific model performs better than the joint model	Distant summary stage shows the worst performance due to lesser instances of training data	[15]
WBCD	Least square—support vector machine (LS-SVM) Hybrid of K-means with SVM (K-SVM)	98.53% 97.38%	- Used a set of linear equations for training; therefore, calculation became easier - Reduces training time by reducing input features from <i>thirty-two to six</i> - Feature selection during training and validation phase not required	Feature Selection not combined with the classification	[48]
	SVM, Decision Tree-C4.5, Naive Bayes, KNN	97.13%	SVM is manifested to be the finest classifier with the least error rate and highest correctly classified instances on WBCD	-	[49]
UCI machine learning repository	SVM, DT-J48, NB, LR	Dermatology—NB + LR (97%) Breast cancer-SVM-PCA (97%) Chronic kidney Disease-J48 (99%)	- High accuracy - Accuracy could be improved by considering more data and parameters such as age, sex, locality - CNN could be used in the future to automate feature extraction.		[50]
					[51]

Sometimes, the results from individual classifiers can be improved by combining several classifiers using different methods, thus producing an ensemble model. In [43], the authors proposed an ensemble framework using four machine learning classifiers—KNN, SVM, DT, and LR and obtained 97.88% accuracy. The sequential least square programming method (SLSQP) was used for assigning weights to each classifier. The findings show that the ensemble model outperforms a standalone machine learning classifier. In Table 4, we discuss another work on SVM ensemble classifiers [12].

Kashif et al. [44] performed image processing, segmentation, and feature extraction on the MIAS dataset using several ML algorithms—SVM, Ada Boost, DT, LR, RF, Gradient Boost, and KNN. Results proved that SVM performs best with an accuracy of 90%. In contrast, the worst performance was given by Gradient Boost classifier with an accuracy of 52%. Another work on the MIAS dataset was proposed in [45], where authors applied discrete wavelet transform (DWT) for feature extraction and PCA for dimension reduction of feature space. The classification was performed using SVM. The combination of DWT, PCA, and SVM approach attains an accuracy of $82.85 \pm 2.21\%$. The proposed work could be continued in the future by applying other variants of wavelength transform, such as stationary wavelet or wavelet package transform. The authors also suggested using deep learning methods on large datasets in the future to obtain high accuracy and precision for prediction and classification.

Cancer prognosis refers to the probability of success of treatment leading to recovery. A lot of work on cancer prognosis has been conducted in the past few years. Similar work was proposed in [46] by evaluating eight different classifiers, firstly, by comparing their AUC value for the prognosis period of 2, 5, 8, and 11 years, and secondly, based on the importance of different variables. The authors concluded that, on average, a 5-year prognosis period shows the overall best performance. Out of eight models, overall RF, Boosted Trees, partial least square, and generalized linear model (GLM Net) classifiers provided the best results with an AUC value of approximately 0.75. In contrast, the KNN model performed worst with an AUC value of 0.67. In the future, their work could be extended by including bagging, boosting, and other ensemble techniques.

Many times, oncologists face difficulties in the identification of the disease through the naked eye. In those cases, radiomics, a branch of medicine, is used to analyze radiological images and thus extract radiomic features to uncover disease characteristics. Sakai et al. [47] classified benign and malignant masses on DBT images using machine learning and radiomic features. Seventy radiomic features relevant to lesion shape, specula existence, and texture information were extracted and fed to different classifiers.

The authors selected six features using the least absolute shrinkage and a selection operator (LASSO) because of lesser training data. Results were evaluated based on the presence and absence of dimension reduction. In the case of 70 radiomic features, SVM shows the best results with a 55% correct recognition rate for a benign tumor and 84% for malignant tumors with an AUC value of 0.798. In contrast, NB attained an AUC value of 0.808 for six radiomic features selected by LASSO. Feature extraction could be automated in the future by implementing deep learning techniques.

Comparative Analysis Using the Strategies of Deep Learning

This section presents the contributions of researchers in recent years for diagnosing breast carcinoma through deep learning techniques.

Al-antari et al. [52] drafted a CAD system on the In-Breast database and implemented the “You Look Only Once (YOLO)” model for detection. The authors proposed a “full resolution convolutional network (Fr-CNN)” and a deep convolutional neural network (DCNN) for segmentation and classification, respectively. Compared with other CAD models, the presented method attains significant performance with a testing time per image of only 12.23 s. The model achieved an accuracy of 95.64% after detection, segmentation, and classification of masses with an AUC value of 0.94. In [53], the authors utilized YOLO for the simultaneous detection and categorization of breast masses on the DDSM dataset. Results show that YOLO achieves remarkable performance with effective mass detection existing over pectoral muscle and dense tissues. Another mass detection model proposed by authors in [54] is based on one stage object detector, i.e., Retina Net. The model gave a higher value of sensitivity w.r.t. False-positive per image (FPPI). The model was implemented on a publicly available dataset, i.e., in breast, and an in-house dataset, i.e., GURO. The authors concluded that performance could be improved by integrating large datasets using transfer learning. Ting et al. [55] utilized an “interactive detection based lesion locator (IDBLL)” and proposed an improved CNN model for breast cancer classification. The authors attained an accuracy value of 90.50% on the MIAS dataset.

In [56], the authors presented an efficient approach for classifying mass lesions on BCDR-FM using a convolutional neural network. The stated framework attains an AUC value of 0.826 and outperforms other state-of-art methods by combining learned and hand-crafted features. Their work could be continued in the future to high-resolution images provided by BCDR-DM. In [57], the authors performed a three-class mammogram classification, i.e., normal, benign, and malignant, on the IRMA dataset by applying a convolutional neural network discrete wavelet (CNN-DW)

and convolutional neural network curvelet transform (CNN-CT). Results have shown that using SVM instead of the Softmax layer can improve classifier performance. CNN-CT outperforms CNN-DW with an accuracy value of 83.74, 83.11, and 81.49 using the SVM layer, SVM layer with tenfold cross-validation, and Softmax layer. Their performance can be enhanced by applying different CNN architectures and various deep learning techniques in the future. H. Li et al. [58] proposed Dense Net II architecture for classifying mammograms. The authors substituted the first convolutional layer with the inception framework. The proposed model shows significant performance when compared with other pre-trained models—Alex Net, VGG Net, Dense Net, and Google Net—and attains an accuracy of 94.55% and AUC of 0.91. Therefore, the model improves speed and efficiency and solves pre-trained models' gradient descent by reducing parameters and enhancing feature re-use. In [59], the authors proposed a supervised approach for evaluating percentage density (PD) and compared the DCNN approach with the statistical learning approach. PD is defined as the proportion of dense area to the breast area on digital mammograms. They concluded that, for PD evaluation, the DCNN method is superior and powerful to a feature-based learning approach. In the future, their system can be extended by including mammograms from the different vendors, using various imaging modalities, and implementing transfer learning.

Pectoral muscles hinder the detection process and should be removed before diagnosis. Mughal et al. [60] proposed a hybrid technique to eliminate pectoral muscle using a curve stitching technique. The adaptive hysteresis thresholding segmentation technique identifies the region of interest (ROI) and segments the breast region of a digital mammogram. Results have shown that this method obtains the loftiest recall of 96.6% on MIAS and 96.4% on the DDSM dataset when compared with other revolutionary methods. As discussed in "Imaging Modalities", FFDM offers low detection sensitivity, and MRI, on the other hand, provides high costs with longer scanning time. "Contrast-enhanced digital mammography (CEDM)" combines the advantages of both, offers four times faster scanning, and is only 1/6 the price. CEDM generates low energy images similar to FFDM and recombined images comparable with breast MRI [61]. For improving the diagnosis of breast carcinoma, "shallow-deep CNN (SD-CNN)" was proposed by authors in [61] by utilizing CEDM. The authors used deep CNN for comparing the results obtained through low energy (LE) image and by the combination of LE and recombined image. Secondly, due to the restricted availability of CEDM, the authors applied shallow CNN (S-CNN) and obtained a 'virtual' recombined image from FFDM images. After evaluation, with virtual recombined imaging features, results show an increase in the model's performance.

Breast calcifications are minute deposits of calcium on the breasts. They commonly appear as white specks or dots on the mammograms. Calcifications are generally non-cancerous, but microcalcification clusters can be an initial symptom of cancer. In [62], the authors performed breast micro-calcification diagnosis using a deep convolutional neural network (DCNN) by utilizing hand-crafted and deep learning features on digital mammograms. They attained 86.89% recall and 89.32% precision using filtered deep features. Researchers concluded that although deep features surpass hand-crafted (morphological) features, on the other hand, hand-crafted features could guide CNN for attaining higher accuracy by providing additional information.

On the other hand, authors in [26] classify multi-view mammograms and segmentation maps using pre-trained deep learning models. As opposed to classifying individual lesions, this holistic approach classifies the whole mammogram containing "Craniocaudal (CC)" and "Mediolateral oblique (MLO)" views. Results show that this semi-automated approach achieves excellent results on In-Breast and DDSM dataset with volume under the ROC surface (VUS) over 0.9 for the three-class problem (normal, benign, malignant) and AUC also over 0.9 for the two-class problem (benign, cancerous). In contrast, a fully automated approach attains VUS and AUC over 0.7 for three-class and two-class problems, respectively.

In medical imaging, multiple instance learning (MIL) provides room for weakly labeled data and reduces the effort and cost of marking all the dataset images. Instead of taking a group of labeled instances, it considers the set of labeled bags comprising multiple instances. P.J. Sudharshan [63] used MIL and implemented several methods on the Break His database. The authors proposed two ways for implementing the process. The first method considers patients as bags and instances are the patches extracted from images, and the second assumes images as bags and instances are the patches. Non-parametric MIL and multiple instance learning convolutional neural network (MILCNN) attain significant results with an accuracy of 92.1%. The authors concluded that MIL outperforms other methods for classifying cancer on histopathology images.

Similarly, Samala et al. [64] concluded that when training samples from one modality are inadequate, multi-task learning is a competent technique for implementing deep networks in medical image processing. Another work on MIL was proposed in [65] for mass detection in DBT images. The authors proposed a framework based on deep CNN (DCNN) with multiple instances learning random forest approach (MILRF), i.e., DCNN-MILRF, and compared the presented structure with two other proposals—first based on hand-crafted features, i.e., MIL-CAD, and second based on deep cardinality restricted Boltzmann machine (deep CaRBM). Results show that the DCNN-MILRF outperforms

the other two approaches with 86.8% accuracy. In the future, their work could be explored with a larger DBT dataset by combining 3D imaging information with 2D information.

In [66], the authors proposed a model for lesion detection on ultrasound images on two datasets, A&B with 306 and 163 images. The authors used three deep learning approaches, patch-based Le-Net & U-Net, and transfer learning-based pre-trained fully convolutional network-AlexNet (FCN-AlexNet). Results show that FCN-AlexNet gave the best performance on dataset A and patch-based Le-Net on dataset B in terms of F-measure and false-positive/image but at the cost of increased training time. Their work could be extended by increasing training data and implementing segmentation and classification on ultrasound lesions. Another similar work on ultrasonographic images was proposed in [67]. The authors used a deep CNN and aimed two frameworks called “Mt-Net” and “Sn-Net” to classify cancerous tumors and solid nodules in the breast lesions. Results reveal that the presented approach surpasses other revolutionary methods and can be elaborated in the future by combining several imaging modalities. Table 5 summarizes some of the well-known research on breast cancer diagnosis using deep learning models.

For training CAD models using deep learning techniques, training data is required in bulk. But the collection of the labeled dataset is a challenging job in the health sector. Different augmentation strategies can be applied to increase dataset size. In recent years, many researchers have used the Break-His dataset to categorize histopathology images due to the widespread availability of a large set of images. In [68, 69], the authors have attained significant performance on classifying breast cancer using histopathology image by comparing the results separately on patch-wise and image-wise classification. In addition to data augmentation, transfer learning can also be utilized when the dataset is not enough for diagnosis. Li Y et al. and Deniz E et al. [70, 71] discussed histology image classification using several pre-trained networks such as Res-Net50, Alex-Net, VGG16, and obtained satisfactory results. In another research, Brancati et al. [72] detected invasive ductal carcinoma. The researchers also performed multi-classification of lymphoma in histological images, using residual convolutional auto-encoder (CAE), i.e., Fusion-Net. Authors implemented two approaches—firstly, classification by reconstruction, i.e., training CAE in an unsupervised manner, and secondly, supervised classification, i.e., guiding only the encoder part of Fusion-Net called Supervised Encoder Fusion-Net (SEF). The authors compared results obtained using SEF with other existing deep learning networks (Fusion-Net, U-Net, Res-Net) under the same conditions and on the same datasets. They concluded that SEF provides outstanding results for the considered cases and can be used in the future for other histopathology image analysis cases.

Jiang et al. [73] proposed a new model, i.e., Breast Cancer Histopathology Image Classification network (BHC Net). The authors reduced training parameters by designing a new module called the small Squeeze and Excitation Residual network (SE-ResNet) and prevented the model from the risk of overfitting. The proposed method attained an accuracy of 98.87 to 99.34% for binary classification and 90.66 to 93.81% for multi-class classification. Their work can be elaborated to study cell duplication and irregular distribution of color in pathological images in the future. Nahid et al. [74] proposed three models for histopathology image classification—CNN, long-short-term-memory (LSTM), and a combination of CNN and LSTM. The authors found unseen structured and statistical information from the data using two unsupervised clustering techniques—K-Means and Mean shift. CNN with Softmax classifier outperforms the other two models with an accuracy of 91%. The authors in [75] presented a hybrid deep learning approach, i.e., multi-modal data fusion by combining structured data from Electronic Medical Record (EMR) with pathological images. Results show significant improvement in diagnostic accuracy by fusing high dimensional and low dimensional data. Sharma and Mehra [29] suggested a framework for the multi-classification of histopathology images on the Break His dataset. The authors, firstly, employed traditional classifiers for performing classification using hand-crafted features. Secondly, transfer learning was implemented, where pre-trained models “VGG16,” “VGG19,” and “ResNet50” were used as feature extractors and as baseline models. Outcomes show that the approach based on pre-trained models as a feature extractor outperforms the baseline and hand-crafted approach. VGG16 model with a linear SVM classifier attained the highest accuracy. Their work could be elaborated further to perform layer-wise fine-tuning of the model and implement an ensemble model with a pre-trained network.

Bevilacqua et al. [28] perform a supervised classification of breast tomosynthesis images by comparing shallow and deeper neural networks. The authors proposed two frameworks: the first framework was based on hand-crafted (morphological/textual) features, i.e., classification by ANN, and the second was a non-neural classifier (i.e., automatic feature extraction by CNN) in which different CNN models (Google Net, ResNet, Alex Net, and VGG Net) were evaluated. Results evinced that the CNN-based model outperforms the ANN approach. In [76], the authors classify mammograms and DBT images on an in-house dataset by evaluating ten different models using pre-trained Alex Net. In this research, 2D mammography outperforms 3D tomosynthesis models in classifying cancer. However, 3D tomosynthesis has proven to be robust in manual cancer detection. The variation in result might be due to memory limitations; as a result of which the authors considered

Table 5 Summarization of distinguished work on breast cancer diagnosis using deep learning techniques

Data set used	Method adopted	Performance	Pros	Cons	Reference
Mammograms (mini-MIAS)	CNN, Sparse Auto-Encoder (SAE) and Stacked Sparse Auto-Encoder(SSAE)	Accuracy: CNN: 97% SAE: 98.5% SSAE: 98.9%	• Best Result by Stacked Sparse Auto-encoder • Helps to identify smaller masses more accurately	-	[9]
Mammograms (labeled and unlabeled)	CNN with Semi-Supervised Learning (SSL) algorithm	Accuracy: 0.8243 AUC: 0.8818 (labeled and unlabeled data)	• Best result obtained using mixed labeled and unlabeled data • Useful when the labeled data is not available in bulk • Performance of the system increases using unlabeled data	-	[82]
MRI images	DCNN	Accuracy: 98.33	• User Independent and time saving as it took only 36 s. to distinguish 59 cancerous regions out of 60 correctly • Reduces unnecessary biopsies • An optimum system for breast MRI	-	[83]
	Inception network, DCNN, gradient boosting tree classifier	Accuracy: 4-class-96.4 2-class-99.5	• Gradient Boosting tree algorithm performs well even in imbalanced training data and limited samples, thus avoiding overfitting. • Improves performance when DCNN is fused with gradient boosting tree	-	[84]
Histopathological images	Pretrained Xception Network	Pre-eminent performance	• Multi-task CNN outperforms single task CNN. • Fine-tuned CNN performs better than CNN trained from scratch • Inclusion of prior knowledge in feature extraction step overcomes intra-class variance problem in pathological images.	-	[85]
SFM, DM, and DBT	Multi-stage transfer learning	ROI-based & View-Based AUC Single-stage: 0.84 and 0.85 Multi-stage: 0.90 and 0.91	• A supplementary phase of transfer learning from a similar related domain provides an improvement in learning.	• Hyper-parameters such as momentum, learning rate, no. of iterations, and the batch size were not optimized due to smaller datasets and less computational cost. • Did not compare performance with conventional feature engineering methods	[86]

Table 5 (continued)

Data set used	Method adopted	Performance	Pros	Cons	Reference
DBT and FFDM	Pretrained VGG16 network	-	- Transfer learning based DBT model outperforms model based on FFDM - Integrating imaging modalities such as FFDM and DBT, increases classification accuracy.	It did not include breast density and other parameters such as breast asymmetry, retraction of mamilla, etc.	[87]
DDSM, BCDR, In-Breast, MIAS	Pretrained VGG16, ResNet50, Inception v3	Inception v3 models perform best on <i>MIAS(test dataset)</i> Accuracy: 98.23 AUC-0.99	- End-to-end learning maximize overall performance - Fine-tuning performs well till two fully connected (FC) layers after that performance degrades - Max- pooled features performed better as compared to fully connected features with or without pre-processing - Fusion classifier outperforms CNN and convolutional CAD classifier	Mammography fails in the case of the dense breast; thus, performance could be further improved by including other modalities-ultrasound, MRI	[88]
FFDM, US, dynamic contrast-enhanced MRI (DCE-MRI)	Pretrained VGG19 model	AUC (Max pooled features) FFDM: 0.81 US: 0.87 MRI: 0.87 AUC (Fusion classifier) – FFDM: 0.86 US: 0.90, MRI: 0.89		-	[89]

the subset of 3D frames, and thus, significant information loss might have occurred in 3D images. For achieving optimal performance, their research could be extended in the future for developing an ensemble classifier integrating 2D mammography with 3D tomosynthesis images. Samala et al. [77] proposed a DCNN model for mass detection in DBT images and showed a comparison between DCNN and feature-based CAD. Unlike other research, where pictures of natural scenes were utilized for pre-training a DCNN, the authors firstly used mammography images for training a DCNN, and then, this pre-trained model is used for mass detection in DBT images using transfer learning. Results have shown that the AUC value increases from 0.81 to 0.90 after transfer learning with DBT. Secondly, DCNN-based CAD surpasses feature-based CAD and thus improves sensitivity from 83 to 91%. Another work on mass detection on DBT images was proposed in [78] using a “Faster Region-Based Convolutional Neural Network (Faster-RCNN).” The authors modified the DCNN proposed in [66] and compared it with the RCNN. Results show that RCNN outperforms DCNN with an AUC value of 0.96 with a reduced false-positive rate. Thus, the false-positive rate is diminished, but at the sake of cost and speed, i.e., RCNN is sluggish and costly as compared with DCNN.

D. Zhang et al. [79] integrated feature selection and feature extraction techniques on gene expression datasets for anticipating clinical outcomes of breast carcinoma. The classification task consists of two phases—unsupervised feature learning, i.e., combining PCA and auto-encoder, and, second, supervised classifier learning, i.e., constructing ensemble classifier based on the Ada-Boost algorithm. Experimental results proved that the proposed method, i.e., PCA-AE-Ada-Boost, gave AUC of 0.714 and accuracy of 0.85 and performed better than other state-of-art methods. Despite its advantages, the authors highlighted some limitations—firstly, difficulty identifying essential features for prediction tasks; secondly, increasing generalization capability of the proposed method by including more publicly available datasets. Deep learning methods can also be utilized for extracting extensive clinical information for breast carcinoma from multiple categories of clinical notes in Chinese [80]. In [80], the authors proposed a system with two constituents—named entity recognition (NER) and relation recognition component. The bi-directional encoder representations from transformers (BERT) language model was fine-tuned for extracting the notion and features from clinical breast cancer documents. Results have shown that fine-tuned BERT outperforms traditional bi-directional long-short-memory-conditional random fields (Bi-LSTM-CRF) algorithms. The method attains precision, recall, and F-measure of 0.927, 0.939, and 0.935 for NER and 0.976, 0.959, and 0.967 for relation extraction, respectively. For information extraction, the proposed approach considered

only 100 patients; therefore, additional data annotation could be done to increase dataset size in the future. Secondly, the proposed model could be enhanced to integrate data from alternative imaging modalities such as computed tomography (CT), MRI, and pathological microscopic images.

Deep learning can also be utilized for nucleus detection and semantic segmentation in human epidermal growth factor receptor-2 (HER2)-colored images of breast carcinoma. In [4], the authors proposed a framework called Her2Net, which consisted of a deep network along with a trapezoidal long short-term memory (TLSTM) structure. Findings suggest that the stated framework is highly efficient and robust in contrast to other recent methods. Generalization error is another critical parameter that needs to be considered when using transfer learning with limited training data. DCNN can memorize the training data due to millions of parameters and trainable weights. Thus, memorization, over-fitting, and a classifier model with enormous specifications may result in generalization error [81]. Samala et al. [81] considered the effect of learning capacity and transfer learning on the generalization error by plotting ROC curves as an evaluation parameter. The noise was introduced by corrupting image labels and the input image. The authors evaluated the correspondence between learning and memorization of deep networks by freezing convolutional layers of pre-trained networks such as Alex-Net and Google-Net. Results show that generalization error increases when training is done with noisy data. The authors also concluded that a proper strategy should be chosen based on available training data and classification tasks to minimize overfitting and improve the model's generalizability when implementing transfer learning.

Research Gaps and Future Directions

Incidences of breast cancer are increasing at a startling rate around the world. In contrast to developed countries, fatality estimates are comparably high in low-wage and middle-wage countries [22]. Early detection and diagnosis play a crucial role and boost long-term survivability [26]. Evaluating medical images using CAD has been confirmed to be the most effective way for early diagnosis and treatment. The symptomatic potentialities of training methods are impeding the degree of personal competence using deep learning. Thus, the CAD paradigm of the "second opinion" tool is now being shifted to a more collaborative utility [27].

In the past few years, artificial intelligence has come with advanced methods for the practical examination of medical images to aid doctors and physicians at varying disease diagnosis levels [19]. Deep learning models learn high-level and low-level features directly from the input and facilitate multi-modality input [26]. The authors widely use deep

learning in most of the studies because it reduces the need for feature engineering, especially in processing redundant data. Secondly, existing deep learning architectures can easily be adapted or modified on new applications [25]. But its computation and storage cost hinder its real-time execution and deployment to hospitals and clinics. Samala et al. [90] perform network pruning using an evolutionary method. It reduces the count of neurons by 87%, count of parameters by 34%, and total multiply-and-add operations by 95% without reducing performance. Thus, the pruned DCNN requires lesser memory and computation for refining an anonymous image and thus increases the model's capability and robustness. Features can be extracted from any layer in DCNN. Still, authors in [90] used the third fully connected layer, and future work is thus required for analyzing the peculiarity of features in various layers.

Experimental results from studies by different authors have proven that breast cancer diagnosis is better using deep learning than conventional machine learning. Deep learning is beneficial for handling complex, heterogeneous, unstructured, and poorly annotated data, but it needs improvement in data integration, interpretability, security, temporal modeling, and incorporating expert knowledge [91]. Despite its advantages, the following are some crucial challenges that need to be addressed when using deep learning for breast cancer diagnosis:

- Less availability of large-scale reliable datasets from expert doctors for breast cancer diagnosis

Deep learning models need a considerable amount of training images. Most of the authors have used publicly available datasets instead of independent datasets. Independent datasets usually do not have many images from expert doctors, so most authors prefer to use publicly available datasets instead of other ones. But the models developed on public datasets are less reliable as these datasets are already pre-processed. Thus, the health industry could be improved if researchers are provided with large annotated datasets from expert doctors [16].

- Less availability of models pre-trained on medical images

As discussed in "Computer-Aided Diagnosis [18]", CNN can be implemented in several strategies: training from scratch, adopting a transfer learning approach, and as a feature extractor. Most of the selected studies have utilized pre-trained models that are trained on an enormous amount of non-medical images. In the second step, fine-tuning is performed to re-train the pre-trained model on medical images. Thus, this process is time-consuming, costly, complicated, and requires a massive set of resources. In recent years, very few researchers have

focused on training a model on medical images and, thus, utilizing these models further for transfer learning. If the dataset is too narrow, i.e., lesser than 1000 images, pre-trained networks experience overfitting issues. Thus, there arises a demand for domain-specific pre-trained models that are specially trained on medical images that would take less time, fewer resources, and are cost-effective.

- Using different imaging modalities and multiple modalities for breast carcinoma classification [16]

Most researchers have used mammography and histopathology images for breast cancer diagnosis, and very few studies have focused on DBT, US, and MRI. Although DBT has given outstanding results in detecting tumors, it is costly, requires a vast amount of data and extra memory capacity for training. Thus, the same patient could be diagnosed using different modalities to enhance the models' classification ability, thus improving efficiency and reliability. In the literature, few studies have utilized multiple modalities [21, 22, 24, 75, 76, 87, 89, 92], and therefore, more research needs to be focused on this area.

- Mammography craniocaudal (CC) and mediolateral oblique (MLO) views

Arefan et al. [93] evaluated the results using CC and MLO views on mammography patients and estimated that the CC view shows superior performance than the MLO view. The authors also observed an increase in AUC value when the two views are used in combination with dense breasts, but the results are reversed in the whole breast region. Thus, additional research is required for evaluating the performance of both views on dense breasts and the entire breast region individually.

- Unsupervised approaches for breast cancer classification [16]

The majority of the selected studies have implemented supervised deep learning, i.e., CNN, that requires a large annotated dataset for breast cancer diagnosis. But in the medical domain, the collection of labeled datasets is challenging, and most of the available datasets are unlabeled. Therefore, more research is needed for developing CAD models that utilize these unlabeled images, which are an essential source of information.

- Stage-specific breast cancer survivability prediction

Kleinlein and Riaño in [94] used data-driven knowledge for predicting 5-year breast cancer survival on the SEER

database on all the BC instances in 1988–2009. The authors compared stage-specific (in situ, localized, regional, distant) and joint predictive models for three distinct machine learning algorithms—naïve Bayes, logistic regression, and decision tree. Authors concluded that summary stage predictive models achieve superior results compared with the joint model; secondly, it is essential to collaborate all the knowledge attained from various stages with its validity time. More research is thus needed for predicting stage-wise and year-wise survival rates for breast cancer patients.

- Integrating non-imaging information with imaging data

Till now, there are very few CAD models that integrate radiomic features with the imaging data. Geras et al. [24] discussed that in the future, many such models must be developed that could combine non-imaging attributes such as analytical information, cancer history, genetic information, along with image data, to provide early diagnosis for women at high risk of cancer.

- Imbalanced dataset

Another challenge for deep neural networks is the class imbalance problem, i.e., the proportion of positive and negative samples. Many available datasets are still imbalanced. The prediction may be biased towards the more common class when models are trained on an imbalanced dataset and have a pessimistic effect on the classifier's behavior [29]. Thus, the unavailability of balanced datasets is another issue that needs to be addressed [25].

- Future of CAD in medicine

Yanase and Triantaphyllou [20] addressed some crucial issues discussing CAD future in medical science. That includes various approaches such as data acquisition and quality estimation, advanced segmentation, feature extraction/selection, trading with big data, and cultivating essential performance evaluation for all developed models.

- Training deep models using smaller datasets in medical applications

Many studies utilize pre-trained models on large image datasets in recent years, but none uses these models on a smaller dataset. As medical datasets are not abundant in numbers, thus, new algorithms should be developed to train deep models on smaller medical image datasets [17].

- Feature extraction from small medical image datasets

Feature extraction enacts a remarkable role in medical image classification. Lai and Deng [10] address two challenges in this regard that need to be considered. Firstly, extract useful features from the small medical image datasets and, secondly, expertly fuse them from different models.

- Generalizability of deeply learned models

Most of the studies discussed in the survey have used small training datasets. Moreover, trained models were not being exposed to meticulous validation on massive testing data that may increase false positives and false negatives due to overfitting and memorizing training data. Data augmentation reduces the risk of overfitting due to small training set by introducing variations or jittering to the original data, thus improving generalizability. But augmenting the training set is not equivalent to independent training samples of comparable size. Therefore, many such models' generalizability to unseen data of new patients is still unknown and is an important parameter to be considered [19].

Conclusion

Breast carcinoma is the premier category of deadliest cancer reported in females. There arises an urgency for early diagnosis and treatment for reducing the mortality rate. Medical image analysis using CAD has cropped up as an essential field for early diagnosis and prognosis in the healthcare domain. Our survey communicates the present state of the art research disseminated in breast cancer using machine learning and deep learning techniques. Although malignancy cannot be proven without biopsy, early carcinoma detection using imaging modalities is an hour of need. Mammography is used as the “gold standard” for breast carcinoma diagnosis due to its widespread availability compared with others. But it is also being criticized because of its low effectiveness for women with dense breasts. Thus, extensive research needs to be addressed by combining various imaging modalities such as ultrasound, MRI, DBT, and mammography to achieve significant results.

This survey also highlights research challenges and future recommendations for the new researchers in this field. Research in machine learning manifested SVM to be the finest classifier for diagnosing breast carcinoma. Deep learning methods also exhibited an outstanding performance for varied evaluation parameters such as accuracy, sensitivity, specificity, F-measure, etc. High-level and low-level features can be learned automatically from the input image using DCNN. Training deep learning models from scratch is time-consuming and

requires extensive resources. As a result, many studies have utilized pre-trained models and attained remarkable performance. Results have proven that deep learning outperforms conventional machine learning in diagnosing breast carcinoma when the dataset is broad. Research gaps from the recent studies depict that practical and scientific research is an urgent necessity for improving healthcare in the long run.

Abbreviations

AI	Artificial intelligence
ML	Machine learning
DL	Deep learning
CAD	Computer-aided diagnosis
ANN	Artificial neural network
CNN	Convolutional neural network
DCNN	Deep convolutional neural network
SFM	Screen film mammography
FFDM	Full-field digital mammography
DBT	Digital breast tomosynthesis
MRI	Magnetic resonance imaging
HP	Histopathology
US	Ultrasound
UCI	University of California Irvine
WBCD	Wisconsin Breast Cancer Dataset
MIAS	Mammography Image Analysis Society
DDSM	Digital Database for Screening Mammography
PCA	Principal component analysis
SVM	Support vector machine
KNN	K-nearest neighbor
LR	Logistic regression
RBF	Radial basis function
NB	Naïve Bayes
DT	Decision tree
GPU	Graphical processing unit
LASSO	Least absolute shrinkage and a selection operator
ROI	Region of interest

Declarations

Ethics approval and informed consent The article does not contain any studies with human participants or animals performed by any of the authors.

Informed consent This is not required as no human participants are involved.

Conflict of Interest The authors declare that they have no conflict of interest.

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